

(1) Field $\mathbb{F}_{53}$ and deck

Pick a deck of $q = 53$ cards (standard 52 plus one joker).

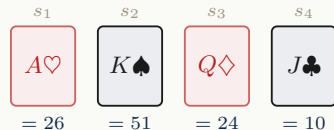
Bijection: $\text{index} = 13 \cdot \text{suit} + \text{rank}$
with $\clubsuit=0, \diamondsuit=1, \heartsuit=2, \spadesuit=3$ and
 $A=0, 2=1, \dots, K=12$.

Sample mappings used below:

$7\clubsuit \rightarrow 6, J\clubsuit \rightarrow 10, 8\diamondsuit \rightarrow 20,$
 $Q\diamondsuit \rightarrow 24, A\heartsuit \rightarrow 26, T\heartsuit \rightarrow 35,$
 $9\spadesuit \rightarrow 47, K\spadesuit \rightarrow 51$

(2) Dealer's 8 secret cards

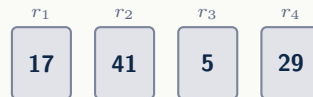
Pick 8 face-down cards. Encode via Step 1 to obtain $\mathbf{s} \in \mathbb{F}_{53}^8$.



$s_5 = 35 (T\heartsuit), s_6 = 47 (9\spadesuit),$
 $s_7 = 20 (8\diamondsuit), s_8 = 6 (7\clubsuit)$

(3) Encoder samples 4 random pad values

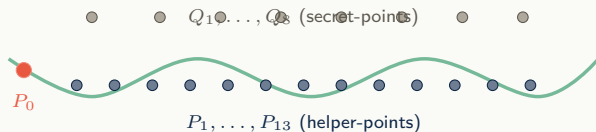
Draw r_1, r_2, r_3, r_4 uniformly from $\mathbb{F}_{53}$:



The pad pins f at the first 4 helper-points:
 $f(P_j) = r_j$ for $j=1, \dots, 4$. It guarantees that
any 4-subset of helper-cards looks uniform.

(4) Pick 22 rational points on Bring's curve $C/\mathbb{F}_{53}$

Choose 22 distinct points in generic position: one anchor P_0 , thirteen helper-points $P_1, \dots, P_{13}$, eight secret-points $Q_1, \dots, Q_8$.



Bring's curve $C \subset \mathbb{P}^4$, $g = 4$; Hasse-Weil gives enough $\mathbb{F}_{53}$ -points.

(5) Encoder solves for $f \in L(15P_0)$

Riemann-Roch: $\dim L(15P_0) = 15 - g + 1 = 12$.

Twelve evaluation constraints pin f uniquely:

- $f(Q_i) = s_i$ for $i = 1, \dots, 8$ (secret)
- $f(P_j) = r_j$ for $j = 1, 2, 3, 4$ (pad)

Solve a 12×12 linear system over $\mathbb{F}_{53}$:

$$f = \sum_{a=1}^8 s_a \phi_a + \sum_{b=1}^4 r_b \psi_b$$

with $\{\phi_a, \psi_b\}$ the basis of $L(15P_0)$ dual to the 12 constraint points.

(6) Distribute 13 helper-cards: $c_j = f(P_j) \in \mathbb{F}_{53}$



Each c_j goes to one player-party (circle below). The first four $c_1, \dots, c_4$ equal the pad $r_1, \dots, r_4$ by construction. The remaining $c_5, \dots, c_{13}$ are linear functionals of $\mathbf{s}$ and $\mathbf{r}$ through f ; the values shown are illustrative (they depend on the chosen $P_5, \dots, P_{13}$ and the basis of $L(15P_0)$).